

Program 4

Step 1: create package.json by running 'npm init -y'

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ ls
Dockerfile index.js node_modules package.json package-lock.json
```

Step 2: Create [index.js](#)

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ cat index.js
const express = require("express");
const app = express;
const PORT = 3000;

app.get("/", (req, res)=> {
  res.send("Hello from Git + docker Integration");
});

app.listen(PORT, ()=>{
  console.log("Server running on https://localhost:`{PORT}`");
})
```

Step 3: Create Dockerfile

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ cat Dockerfile
#Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json /.
RUN npm install
COPY . .

#Deploy Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["nodejs", "index.js"]
```

Step 4: Build Docker image

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ docker build -t program4:latest .
[+] Building 114.1s (14/14) FINISHED
```

Step 5: Check for Docker image

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
program4            latest      e0f6a2772fc0     About a minute ago 195MB
```

Step 6: Initialize git repository and commit the code

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ git add .
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ git commit -m "Initail Commit"
[master (root-commit) c0110cd] Initail Commit
636 files changed, 72889 insertions(+)
```

Step 7: Initialize remote repository and push into it

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ git remote add origin https://github.com/chandan-athreya-s/docker-program4.git
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog4$ git push -u origin main
Username for 'https://github.com': chandan-athreya-s
Password for 'https://chandan-athreya-s@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 8 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 724.88 KiB | 7.40 MiB/s, done.
Total 719 (delta 140), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (140/140), done.
To https://github.com/chandan-athreya-s/docker-program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```